



Research Article

Identification of Critical Water-Quality Parameters in an Abattoir-Impacted River Using Principal Component and Multivariate Statistical Analyses

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Abstract

Abattoir wastewater introduces organic matter, nutrients, suspended solids, microorganisms and potentially toxic metals into receiving rivers, producing complex water-quality responses that are difficult to interpret using single-parameter assessments. This study identified the critical variables and dominant pollution processes controlling water quality in the Effurun River, Delta State, Nigeria, using principal component analysis (PCA) and multivariate statistical analysis. Water samples were collected at an upstream control station, the abattoir-effluent discharge point and a downstream recovery station during rainy-season (June 2025) and dry-season (November 2025) campaigns on Days 1, 15 and 30, yielding 18 station-level observations. Physicochemical, nutrient, organic, heavy-metal and microbiological parameters were determined using standard laboratory procedures. Seasonal differences in the combined water-quality profile were assessed by multivariate analysis of variance, while PCA with varimax rotation was used for dimensionality reduction and identification of variables with absolute loadings of at least 0.70. Four retained principal components explained 92.62% of the total variance. After rotation, PC1 explained 49.12% and represented ionic-organic enrichment dominated by electrical conductivity, total dissolved solids, major ions, nutrients, chemical and biochemical oxygen demand, ammonium and heterotrophic bacteria. PC2 explained 25.48% and represented heavy-metal contamination, with high loadings for Pb, Zn, Cu, Fe, Cd, Mn and Cr. PC3 explained 12.15% and reflected suspended-solids, nutrient and microbial pollution, while PC4 explained 5.87% and represented oxygen-regime processes dominated by dissolved oxygen. The seasonal MANOVA was not statistically significant (Pillai's Trace = 0.667, $F(14, 3) = 0.429$, $p = 0.881$; partial eta squared = 0.667), indicating that the overall seasonal difference was not established at $\alpha = 0.05$. The findings show that a relatively small group of highly loaded parameters can serve as priority indicators for routine monitoring and targeted pollution control in abattoir-impacted tropical rivers.

Keywords

Abattoir Effluent, Principal Component Analysis, Multivariate Analysis, Critical Water-quality Parameters, Heavy Metals, Effurun River, Pollution Source Identification

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Received: 27 August 2026; Accepted: 7 September 2026; Published: 30 September 2026



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1. Introduction

Surface-water systems in rapidly urbanizing tropical environments receive multiple contaminant inputs from domestic wastewater, solid-waste disposal, storm runoff and small-scale industrial activities. Abattoirs are particularly important point sources because slaughtering and wash-down operations generate wastewater containing blood, animal tissue, fats, faecal matter, nutrients, suspended solids and microorganisms. When discharged without adequate treatment, these constituents increase oxygen demand, alter ionic composition and nutrient status, and may introduce toxic metals and pathogens into receiving waters [1-4]. Conventional water-quality assessment commonly evaluates individual variables against guideline values. Although necessary for regulatory interpretation, this approach can become difficult when many strongly correlated parameters are measured simultaneously.

Multivariate statistical methods provide a complementary framework by reducing redundancy, identifying correlated groups of variables and revealing the underlying processes responsible for observed water-quality variation [5-8]. Recent applications in Nigerian rivers and coastal waters have demonstrated the usefulness of principal component analysis (PCA) for identifying dominant pollutant groups, distinguishing hydrochemical processes and reducing complex water-quality datasets to a smaller number of diagnostically useful variables [17-19]. Principal component analysis (PCA), in particular, transforms a large set of interrelated variables into a smaller number of orthogonal components, thereby facilitating identification of critical indicators and potential pollution-source signatures. The Effurun River in Delta State, Nigeria, receives untreated wastewater from an abattoir located close to the river bank. The underlying project measured physicochemical, nutrient, organic, heavy-metal and microbiological characteristics at upstream, discharge-point and downstream stations in contrasting wet and dry seasons. Earlier project-level interpretation showed pronounced spatial and seasonal deterioration, but effective monitoring requires more than demonstrating that pollution exists; it requires determining which parameters carry the greatest diagnostic information and what dominant processes they represent.

Accordingly, this study aimed to identify the critical water-quality parameters controlling degradation of the Effurun River using PCA with varimax rotation and multivariate seasonal analysis. The objectives were to: (i) determine whether the combined water-quality profile differed between seasons; (ii) reduce the multidimensional dataset to a limited number of dominant principal components; (iii) classify high-loading variables according to the major pollution processes they represent; and (iv) derive a priority set of indicators for routine monitoring of abattoir-impacted river reaches.

2. Materials and Methods

2.1. Study Area and Abattoir Setting

The study was conducted on the Effurun River in Uvwie Local Government Area, Delta State, southern Nigeria. The river is situated in the Niger Delta and is exposed to residential, commercial and industrial pressures. The investigated abattoir is a small-scale slaughter facility with a concrete slaughter slab and perimeter drainage. Process water and blood-containing wash water are discharged through drainage channels into the river. The river supports domestic, agricultural and recreational activities in surrounding communities.

2.2. Sampling Design

A seasonal comparative design was adopted. Three georeferenced sampling stations were selected to represent background conditions, immediate effluent impact and downstream transport/recovery. Sampling was undertaken during the rainy season in June 2025 and the dry season in November 2025 on Days 1, 15 and 30 of each seasonal campaign. Samples were collected at all three stations on each occasion, producing 18 station-level observations, excluding analytical replicates and quality-control samples.

Table 1. Sampling stations used for the multivariate analysis.

Station	Functional description	Coordinates
A	Upstream control/reference	5.555282 N, 5.7864476 E
B	Abattoir-effluent discharge/impact zone	5.555384 N, 5.7864677 E
C	Downstream recovery zone (~100 m)	5.5552787 N, 5.786777 E

Samples for physicochemical and microbiological analysis were collected in clean 1-L polyethylene bottles. Bacteriological bottles were sterilized before use, while bottles for physicochemical analysis were rinsed at the sampling site. Samples were labeled, cooled to approximately 4 °C and transported for laboratory analysis. Field measurements were conducted within a standardized morning sampling window to improve comparability among stations and seasons.

2.3. Water-Quality Determinations and Quality Assurance

The analytical programme included pH, electrical conductivity (EC), turbidity, dissolved oxygen (DO), total dissolved solids (TDS), total suspended solids (TSS), biochemical oxygen demand (BOD), chemical oxygen demand (COD), major

ions and nutrients, selected heavy metals and microbiological indicators. Heavy metals included Fe, Zn, Cu, Pb, Cd, Mn and Cr in the multivariate dataset. Microbial indicators included total heterotrophic bacterial count (THBC) and total coliforms. Standard analytical procedures were applied, with daily calibration of instruments, analytical blanks and duplicates, certified or standard solutions for metal analysis, and sterile procedures for microbiological testing [1, 9] and [12-14].

2.4. Multivariate Statistical Analysis

Descriptive statistics were first used to characterize seasonal variability. A multivariate analysis of variance (MANOVA) was then used to test whether the combined water-quality profile differed between dry- and rainy-season groups at $\alpha = 0.05$. The project output reported Pillai's Trace, Wilks' Lambda, Hotelling's Trace and Roy's Largest Root. An internal consistency check of the reported seasonal statistics showed that $F(14, 3) = 0.429$ corresponds to p approximately 0.881. The calculated probability value is also consistent with the project-reported noncentrality parameter of 6.000 and observed power of 0.081. Therefore, $p = 0.881$ was used for the seasonal MANOVA interpretation, while partial eta squared = 0.667 was retained as reported.

PCA was applied to the standardized multivariate dataset and rotated using the varimax criterion to improve interpreta-

bility. Components with eigenvalues greater than 1 were retained in accordance with the Kaiser criterion used in the project analysis. Parameters with absolute rotated loadings of at least 0.70 were treated as critical variables for component interpretation. Communalities were examined to assess how effectively the retained component solution represented the original variables. Statistical analyses were performed using SPSS Version 27.0 and Microsoft Excel 2021.

3. Results

3.1. Multivariate Seasonal Effect

The multivariate test did not show a statistically significant seasonal difference in the combined water-quality profile. For the seasonal grouping effect, the project output reported Pillai's Trace = 0.667, Wilks' Lambda = 0.333, Hotelling's Trace = 2.000 and Roy's Largest Root = 2.000, with $F(14, 3) = 0.429$. The probability value corresponding to this F statistic and its degrees of freedom is $p = 0.881$; therefore, the null hypothesis of no overall seasonal difference is not rejected at $\alpha = 0.05$. The reported partial eta squared of 0.667 is numerically large, but it should be interpreted cautiously because the analysis had only 3 error degrees of freedom and low observed power (0.081).

Table 2. Multivariate test statistics for the seasonal grouping effect.

Statistic	Value	F	Hypothesis df	Error df	p	Partial eta ²
Pillai's Trace	0.667	0.429	14	3	0.881	0.667
Wilks' Lambda	0.333	0.429	14	3	0.881	0.667
Hotelling's Trace	2.000	0.429	14	3	0.881	0.667
Roy's Largest Root	2.000	0.429	14	3	0.881	0.667

3.2. PCA Dimensionality and Variance Explained

Four principal components had eigenvalues greater than

one and together explained 92.62% of the total variance. Before rotation, PC1 had an eigenvalue of 16.027 and explained 53.42% of variance; PC2 had an eigenvalue of 7.846 and explained 26.15%. PC3 and PC4 explained 8.52% and 4.52%, respectively. Varimax rotation redistributed the explained variance to 49.12%, 25.48%, 12.15% and 5.87% for PC1-PC4 without changing the cumulative variance.

Table 3. Variance explained by the four retained principal components.

Component	Initial eigenvalue	Initial variance (%)	Rotated variance (%)	Cumulative rotated variance (%)
PC1	16.027	53.422	49.115	49.115
PC2	7.846	26.152	25.481	74.597

Component	Initial eigenvalue	Initial variance (%)	Rotated variance (%)	Cumulative rotated variance (%)
PC3	2.557	8.523	12.146	86.743
PC4	1.356	4.518	5.873	92.616

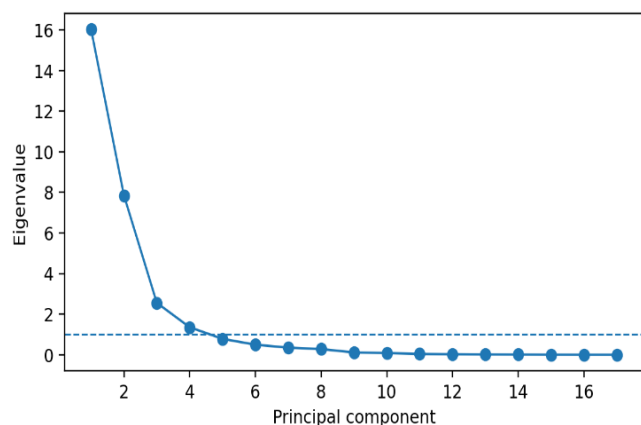


Figure 1. Scree plot reconstructed from the eigenvalues reported in the project PCA output.

The scree profile showed a steep reduction after PC1 and PC2 and an elbow around PC4, supporting retention of four components. The large first eigenvalue demonstrates that the water-quality dataset was dominated by a strong common pollution/hydro chemical signal rather than many equally important independent processes.

3.3. Rotated Component Structure and Critical Parameters

The rotated component matrix revealed four interpretable pollution-process groups. PC1 was dominated by dissolved-ion, nutrient, organic-load and microbial variables; PC2 by heavy metals and acid-alkaline control; PC3 by suspended solids, nitrate and coliforms; and PC4 by dissolved oxygen. Using $|\text{loading}| \geq 0.70$ as the project threshold, the most diagnostic variables are summarized in Table 4.

Table 4. Classification of critical water-quality parameters based on rotated PCA loadings.

Component	Dominant interpretation	Rotated variance (%)	Critical parameters ($ \text{loading} \geq 0.70$)
PC1	Ionic-organic enrichment / mineralization and abattoir-related organic loading	49.12	EC, TDS, Na, SO ₄ , Cl, HCO ₃ , NO ₂ , Mg, P, Ca, salinity, COD, BOD, K, NH ₄ -N, THBC
PC2	Heavy-metal contamination and acid-alkaline control	25.48	Pb, Zn, Cu, Fe, Cd, Mn, Cr, pH (-), alkalinity (-)
PC3	Suspended-solids, nutrient and microbial pollution	12.15	Turbidity, TSS, NO ₃ , total coliforms
PC4	Oxygen regime and biochemical stabilization	5.87	DO

Selected high rotated loadings illustrate the separation of the four factors: Na (0.987), COD (0.988), Ca (0.984), phosphate (0.979), Mg (0.972), NO₂ (0.964) and Cl (0.953) loaded strongly on PC1; Zn (0.934), Mn (0.930), Pb (0.919), Cu (0.919), Cd (0.887), Fe (0.862) and Cr (0.804) loaded on PC2; TSS (0.907), turbidity (0.903) and total coliforms (0.850) loaded on PC3; and DO (0.804) dominated PC4.

The strongest PC1 loadings were observed for COD (0.988), Na (0.987), Ca (0.984), phosphate (0.979), Mg (0.972), K (0.968), NO₂ (0.964), Cl (0.953), HCO₃ (0.947), NH₄-N (0.943), SO₄ (0.935), TDS (0.903), EC (0.866), salinity (0.839), BOD (0.809) and THBC (0.703). This concentration of high loadings across dissolved ions, nutrients, organic-load indicators and microbial abundance supports interpretation of

PC1 as a combined mineralization and organic-effluent factor. The result also shows why no single chemical variable is sufficient to characterize the pollution process: several parameter classes respond simultaneously to wastewater input and river-water chemistry.

For PC2, the strongest positive loadings were Zn (0.934), Mn (0.930), Pb (0.919), Cu (0.919), Cd (0.887), Fe (0.862) and Cr (0.804), while pH (-0.771) and alkalinity (-0.739) loaded negatively. This pattern indicates a distinct metal-control factor with an inverse acid-alkaline relationship. For PC3, TSS (0.907), turbidity (0.903) and total coliforms (0.850) were the clearest high-loading variables. The project classification table also lists nitrate under PC3, although its reported

rotated loading is 0.627, below the stated $|\text{loading}| \geq 0.70$ criterion; this discrepancy should be checked against the original SPSS output before submission. DO (0.804) dominated PC4 and therefore provided the most direct indicator of the oxygen-regime factor.

The loading pattern has practical monitoring value because it permits representative indicators to be selected from each factor. Where resources are limited, monitoring programmes may use one or more high-loading variables from each component as screening indicators, followed by full laboratory analysis when deterioration is detected. Such reduction should be treated as a surveillance strategy rather than a replacement for statutory compliance testing, especially where toxic metals or microbiological hazards are involved.

3.4. Critical-Parameter Monitoring Set

The PCA results indicate that routine monitoring can be made more diagnostic by prioritizing parameters that represent each major process rather than treating all measured variables as equally informative. A practical core set should include EC or TDS for ionic enrichment; COD and BOD for organic loading; $\text{NH}_4\text{-N}$ or NO_2 and phosphate for nutrient contamination; Pb and Cd as high-concern toxic metals together with a broader metal screen where resources permit; turbidity/TSS for particulate contamination; total coliforms or THBC for microbial pollution; and DO for ecological oxygen stress. This reduced set does not replace full regulatory monitoring, but it provides a statistically justified screening panel for detecting deterioration and triggering more comprehensive analysis.

4. Discussion

4.1. Dominant Ionic-Organic Pollution Factor

PC1 explained nearly half of the rotated variance, making ionic-organic enrichment the dominant process in the dataset. The simultaneous loading of EC, TDS, major cations and anions, nutrients, COD, BOD, ammonium and THBC indicates that mineralization and organic contamination were not independent phenomena. This pattern is consistent with wastewater containing dissolved salts, blood residues, proteins, fats, animal excreta and wash water. Such effluent increases dissolved ionic strength and oxygen demand while supporting microbial growth. Similar multidimensional deterioration has been reported in rivers receiving untreated slaughterhouse and mixed urban effluents [2-4, 6].

The strong loadings of COD and BOD are especially important because they directly characterize the oxygen demand imposed by oxidizable and biodegradable organic matter. Their association with ammonium, phosphate and THBC suggests a coupled organic-nutrient-microbial process rather than isolated exceedances. EC and TDS can therefore function as

rapid field indicators of changes in the broader ionic component, but they should be interpreted alongside oxygen-demand and nutrient indicators to avoid attributing natural mineralization solely to abattoir activity.

4.2. Heavy-Metal Contamination Factor

The grouping of Pb, Zn, Cu, Fe, Cd, Mn and Cr in the present study is also consistent with recent investigations of pollution associated with abattoir activities in Nigeria. Elevated concentrations of several heavy metals have been reported in slaughterhouse-affected environments, although the magnitude and specific metal profiles vary among locations and receiving environments [16, 20]. These findings reinforce the importance of retaining selected metals as sentinel variables in monitoring programmes. The public-health significance of this component is substantial because Pb and Cd are toxic at low concentrations and may accumulate through prolonged exposure. Their high loadings justify inclusion in any targeted monitoring programme even when broader routine chemistry is reduced [7, 8, 10].

4.3. Suspended-Solids, Nutrient and Microbial Factor

PC3 grouped turbidity, TSS, nitrate and total coliforms. The association of particulate material with microbial indicators suggests simultaneous transport of suspended organic matter and faecal contamination. In an abattoir setting, this pattern consists of animal waste, undigested feed, soil particles and contaminated wash water entering the river, particularly during rainfall and surface runoff. This component therefore represents both an operational waste-management problem and a catchment-runoff problem. Recent evidence from Nigerian streams receiving abattoir effluents also demonstrates the importance of microbial indicators in assessing receiving-water impairment. [15, 20] reported substantial heterotrophic bacterial and coliform loads in Yewa and Iju streams receiving abattoir effluent, together with associated physicochemical and heavy-metal contamination.

4.4. Oxygen-Regime Factor and Seasonal Modulation

DO formed a distinct fourth component, reflecting the balance among oxygen consumption, reaeration, temperature, flow and photosynthetic activity. Its separation from the main organic-load component indicates that oxygen conditions were influenced by hydrological and ecological processes in addition to pollutant loading. Although descriptive seasonal contrasts were observed, the MANOVA did not establish a statistically significant overall seasonal effect ($F(14, 3) = 0.429$, $p = 0.881$). The numerically large partial eta squared (0.667) should be interpreted cautiously in view of the low

observed power (0.081) and the small residual degrees of freedom. Seasonal flow conditions may nevertheless alter dilution, pollutant residence time, sediment suspension and catchment runoff, thereby changing individual water-quality variables even when the overall multivariate seasonal effect is not statistically significant [4, 11].

4.5. Implications for Monitoring and Pollution Control

The main management value of the PCA is its ability to translate a large laboratory dataset into four interpretable process groups. For routine surveillance, the highest-loading variables provide a rational basis for selecting sentinel parameters. EC/TDS, COD/BOD, selected nutrients, Pb/Cd, turbidity/TSS, microbial indicators and DO collectively cover the dominant ionic, organic, metal, particulate, microbial and ecological dimensions identified by the analysis. Monitoring these indicators upstream, at the discharge point and downstream would provide a sensitive early-warning framework for detecting changes in abattoir-effluent impact. Pollution control should prioritize treatment at source. Screening and solids removal, separation of blood and fats, biological treatment for biodegradable organic matter, nutrient reduction and appropriate disinfection would directly address the major PCA factors. Because the heavy-metal component may include both abattoir-related and catchment sources, source control should be complemented by broader urban-runoff and waste-management interventions.

4.6. Study Limitations

The analysis is based on two intensive one-month seasonal campaigns with three sampling dates per season and three river stations, giving 18 station-level observations. It therefore characterizes seasonal contrasts but not continuous year-round dynamics. With 14 dependent variables in the reported seasonal MANOVA, only 3 error degrees of freedom remained, and the observed power was low (0.081); this limits the precision and sensitivity of the multivariate seasonal inference. The $p = 0.002$ value in the project table was internally inconsistent with $F(14, 3) = 0.429$ and was corrected to $p = 0.881$ for this manuscript. The project output available for this manuscript did not report Kaiser-Meyer-Olkin or Bartlett's sphericity statistics; these diagnostics should be added if available from the original SPSS analysis. Finally, PCA identifies correlation structure and likely common processes; it should not be interpreted as definitive proof that every variable within a component has a single pollution source.

5. Conclusions

Principal component and multivariate statistical analyses identified a compact set of processes controlling water-quality deterioration in the Effurun River. Four rotated components

explained 92.62% of total variance. The dominant component (49.12%) represented ionic-organic enrichment and included EC, TDS, major ions, nutrients, COD, BOD, ammonium and heterotrophic bacteria. The second component (25.48%) represented a distinct heavy-metal factor dominated by Pb, Zn, Cu, Fe, Cd, Mn and Cr. The third component (12.15%) represented suspended-solids, nutrient and microbial pollution, while the fourth component (5.87%) represented oxygen-regime processes dominated by DO.

The multivariate seasonal analysis did not provide statistically significant evidence that the collective water-quality profile differed between rainy and dry seasons (Pillai's Trace = 0.667, $F(14, 3) = 0.429$, $p = 0.881$; partial eta squared = 0.667). Descriptive seasonal differences were nevertheless observed, but the low observed power (0.081) requires cautious interpretation of the seasonal MANOVA. Together with the PCA results, the findings demonstrate that abattoir-impacted river monitoring can be strengthened by focusing on high-loading sentinel variables that represent each dominant process. A priority monitoring panel comprising EC/TDS, COD/BOD, selected nutrients, Pb/Cd, turbidity/TSS, microbial indicators and DO would provide a practical screening framework, while full regulatory analysis should be retained for compliance and health-risk assessment.

Effective restoration requires source-level abattoir wastewater treatment combined with routine upstream-discharge-downstream monitoring and broader control of catchment runoff and waste disposal. Before journal submission, the original SPSS MANOVA output should be verified and KMO/Bartlett diagnostics should be added if available.

Abbreviations

PCA	Principal Component Analysis
MANOVA	Multivariate Analysis of Variance
PC	Principal Component
EC	Electrical Conductivity
TDS	Total Dissolved Solids
TSS	Total Suspended Solids
DO	Dissolved Oxygen
BOD	Biochemical Oxygen Demand
COD	Chemical Oxygen Demand
THBC	Total Heterotrophic Bacterial Count
SPSS	Statistical Package for the Social Sciences
KMO	Kaiser–Meyer–Olkin
APHA	American Public Health Association
WHO	World Health Organization
NESREA	National Environmental Standards and Regulations Enforcement Agency

Acknowledgments

The author gratefully acknowledges Engr. Prof. Hilary Owamah for research supervision and scholarly guidance, and

the laboratory personnel who provided technical support during the water-quality analyses

Author Contributions

Onyeka Edike: Conceptualisation, Data curation, Formal Analysis, Investigation, Methodology, Visualisation, Writing – original draft

Emmanuel Ebikabowei Enemugha: Writing – review & editing

Funding

The underlying postgraduate research was supported by the Tertiary Education Trust Fund (TETFund). A specific grant number was not stated in the project document.

Data Availability Statement

The data supporting the findings are contained in the underlying postgraduate research dataset and may be made available by the corresponding author subject to institutional and ethical requirements.

Conflicts of Interest

The author declares no conflicts of interest.

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Biography



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